



Questions Lab Micro

Mid Term

Healing Group



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Questions lab micro (Mid Term)

Q) you believe you have isolated a species of Bacillus, and an isolated colony is on a plate which was incubated for two days, what portion of the colony would you use to get a correct classification of this bacteria through Endospore staining:

(C) Edge if the colony

(C) Both center and edge of the colony can be used

(C+) center of the colony

(C) the part which is in between the edge and the center of the colony

(C) all the choices are incorrect

Q) The bacteria that make a mucoid mixture with 10 KOH would have which one of the following microscopic appearances immediately after decolorizing steps of Gram staining?

(C) blue/violet

(C) red/pink

(C) yellow

(C+) colorless

(C) green

Q] What type of staining procedure is the Gram staining?

(C) Negative

(C+) Differential

(C) Acid-fast

(C) Simple

(C) Structural

Q) If a complete gram staining was done to a mixed side of Bacillus, Mycobacterium and E. coli Which one of these appear colored under the microscope?

(C) Bacillus

(C) E. coli

(C) mycobacterium

(C+) Both Bacillus and E. coli

(C) Both Bacillus and mycobacterium

Q) In Gram staining the washing step after addition of mordant is followed by:

(C) Loading the slide with crystal violet

(C) Loading the slide with safranin

(C+) Decolorizing step

(C) Loading the slide with iodine

(C) Loading the slide with methylene blue

Q) At the same time, you prepared three bacterial slides from a 20 hours old pure culture through Gram staining, All of the following will explain the variability between these slides if you use the same Gram staining solution set EXCEPT :

(C) Thickness of the smear

(C+) Freshness of the reagents.

(C) Time of decolorizing process

(C) Thoroughness of washing by tap water.

(C) Time of heat fixation

Q) Gram positive bacteria will appear _____ at the end of Gram staining :

- (C) Colorless
 - (C) Colorless at the center and pink cell wall
 - (C) green
 - (C+) violet / blue**
 - (C) pink / red
-

Q) Bacteria that mostly grow on MacConkey well appear _____ at the end of Gram staining:

- (C) Colorless
 - (C) Colorless at the center and pink cell wall
 - (C) green
 - (C) violet / blue
 - (C+) pink / red**
-

Q) In dilution, one of the following is true about the original volume:

- (C) Number of bacteria in it must be equal to that in same volume of the source
 - (C) contains the microorganisms to be diluted
 - (C) Determine the inoculum
 - (C) it is the volume which we add
 - (C+) usually free of microorganisms**
-

Q) One of the following NOT used to do the surface spreading method:

- (C) Turntable
- (C+) inoculating loop**
- (C) Test tube shaker
- (C) Test tubes
- (C) Bottle

Q) A student inoculated 0.1 ml from 10^5 several times but in times he got uncountable plate. This student must be:

- (C) Use a lower dilution for inoculation
- (C) Change the substance he used as inoculum

(C+) Use a higher dilution for inoculation

- (C) Use other type of media
 - (C) Only repeat the inoculation he gets countable plates
-

Q) All of the following are used to prepare a wet mount slide of bacteria EXCEPT :

- (C) Sides
- (C) Sterile inoculating loop
- (C) Coverslip

(C+) Crystal violet

- (C) Sterile distilled water
-

Q) Structural staining use one or more dyes to stain certain structure within bacteria such as Endospores, Capsule, flagella and Genetic material :

(C+) True

- (C) False
 - (C) The information is insufficient
-

Q) Which is common the Gram staining and endospore staining:

- (C) use of iodine as mordant
- (C) use of steam for stain penetration.

(C+) use of Safranin as a counter stain

- (C) Fixation of the smear prior to Staining.
- (C) use of acid alcohol for decolorizing

Q) While you begin heating a growing troth culture. The fire alarm rings and you must leave the building quickly, when you return you discover the medium is completely boiled away and only ashes remain in the bottom of the culture tube, In desperation to retrieve your culture you add sterile medium to the culture tube and incubate at 37 Covernight. The next day you discover your culture growing very well.

The subsequent experiments show that this growth is NOT a contaminant the most important microscopic characteristic of this bacteria is:

(C+) Spore former bacteria

- (C) Acid-Fast bacteria
 - (C) Hemophilic bacteria
 - (C) Gram positive bacteria
 - (C) Gram negative bacteria
-

Q) When E.coli and Bacillus is incubated on a pure dish of phenol ethanol agar. Only E. coli grew. In this respect phenol ethanol agar is an example of:

- (C) A general-purpose medium

(C+) A selective medium

- (C) differential medium
 - (C) complex medium
-

Q) Which one the following statements is false about Bacillus cereus?

- (C) Is Gram positive bacillus

(C+) can grow on MacConkey agar

- (C) Can stained by safranin
 - (C) Spore Former
 - (C) non acid -fast Bacteria
-

Q) All of the following are True about checking the cooling of the loop by entering it into media away from bacteria EXCEPT:

- (C) Prevent killing of bacteria.
- (C) Prevent scattering of bacteria in the lab air.

(C+) Prevent contamination of bacterial culture

- (C) prevent pollution of the indoor lab air
 - (C) prevent our cell from infection
-

Q) If a bacterial culture was broken at your tabletop which Of the following you should Do :

- (C) cover it with tissue paper then discard it at the end of your microbiological work into the miter container
- (C) Cover it with tissue paper then saturate It with disinfectant and discard it immediately into designated container

(C+) cover it with tissue paper then saturate it with 70% alcohol and discard it after 15 minutes into designated container.

- (C) Cover It with Tissue paper then situate It with 95% alcohol and discard It after 15 minutes into designated container
 - (C) Cover it with tissue paper then saturate a with 70% alcohol and discard it immediately into designated container
-

Q) After you examined the bacterial slides under the microscope, you must:

(C+) discard the slides immediately in a disinfectant container

- (C) Leave on the table top until you finish your microbiological work then discard all bacterial slides in a disinfectant container.
- (C) Discard slide in a garbage container
- (C) You can wash the slide with lap water then reuse it

Q) One the following is false during your microbiological work:

(C+) Drying of bacterial smear can be achieved by blowing it in air

- (C) Never remove bacterial culture from the laboratory.
 - (C) During aseptic transfer OF bacteria on your table top you must close the air currents opening the Lab
 - (C) Eating ,drinking and chewing is prevented during your microbiological work
-

Q) which one of the following is reason for tabletop disinfecting before your microbiological work but NOT for closing the air currents:

- (C) prevent pollution of the outdoor air
 - (C) prevent contamination of our culture
 - (C+) prevent infection of our self**
 - (C) prevent pollution indoor air
-

Q) Bacterial cells have an overall negative charge on their surface. If a bacterial smear is stained by a pink, acidic, water insoluble stain and then washed with water , which of the following describes the result :

- (C) pink cells in a pink background
- (C) unstained cells with no background
- (C) pink cells
- (C) purple cells

(C+) unstained cells in pink background

Q) In a negative staining procedure, the bacterial cells would be:

- (C) Stained in an unstained background
- (C) Brightly colored a contrasting background
- (C) Attracted to the dye used in the staining procedure
- (C) The same color as the background and thus not easily observed

(C+) Unstained in a Colored background

Q) The water used in wet mount preparation is:

(C+) Sterile distilled water

- (C) Sterile deionized water
 - (C) Tap Water
 - (C) Any type of sterile water
 - (C) Any type of distilled water
-

Q) Wet mount can be used to:

- (C) Differentiate between gram positive and gram-negative bacteria.
- (C) Differentiate between Acid fast and non-acid fast bacteria

(C+) Differentiate between motile and non-motile bacteria

- (C) Visualize the Flagella
 - (C) Visualize the capsule
-

Q) Which of the following is correct in the smear preparation:

(C+) Thick smear is not suitable for examining the bacteria.

- (C) The smear is fixed with heat by placing the slide over boiling water for 5 minutes
- (C) The smear is dried by passing the side over the flame
- (C) After heat fixation, the smear must be washed before staining to cool the slide
- (C) Heat fixation is used to fix the size of bacterial cells so it will not change during the staining

Q) Which of the Following is NOT accomplished by fixing cells to a slide:

(C) Cell structures such as the capsule are disturbed

(C) Cells are attached to the side

(C) Cells are killed

(C+) cell structures are differentiated

(C) cell structures are preserved

Q) one of the followings is miss - matched:

(C) Direct staining - one dye

(C+) indirect staining - two dye

(C) wet mount -no dye

(C) gram staining -two dye

(C) Acid fast staining -Two dye

Q) Which of the following stains is used for visualizing Mycobacterium:

(C+) The acid-fast stain

(C) The endospore stains

(C) The Gram stains

(C) The negative stain

Q) In a differential stain:

(C) The slide must never contain more than two different species of bacteria

(C)The positively charged stain colors the Negatively charged cells

(C+) It is possible to distinguish between different types of cells based on how they take up stain

(C) Heat fixation is not needed

(C)The positively charged cell wall attracts the positively charged stain

(C) the positively charged stain colors the negatively charged cells

Q) In a negative stain:

(C+) heat fixation is not needed

- (C) The slide must never contain more than two different species of bacteria
 - (C) it is possible to distinguish between different types of cells based on how they take up stain
 - (C) The positively charged stain colors the negatively charged cells
 - (C) The positively charged cell wall attracts the positively charged stain
-

Q) A cation:

(C) is negatively charged bacteria

(C+) A positively charged ion

- (C) A negatively charged ion
 - (C) A neutrally charged ion
 - (C) A positively charged bacteria
-

Q) Which of the following statements about Gram staining is TRUE?

- (C) Gram staining differentiates bacteria based on the plasma membrane composition
- (C) The Gram stain makes all cells appear red
- (C) All bacteria are either Gram positive or Gram negative

(C+) Some bacteria do not have a cell wall. Therefore, they cannot be reliably Gram stained

- (C) The Gram stain makes all cells appear purple

Q) A mordant is:

- (C) A substance that decolorizes the slide
- (C) The primary purple stain

(C+) A substance that enhances staining ability

- (C) The red counter stains
 - (C) 95% ethanol
-

Q) In dilution, one of the following is true about the original volume:

- (C) Number of bacteria in it must be equal to that in the same volume of the source
- (C) Contains the microorganisms to be diluted
- (C) Determine the inoculum
- (C) is the volume which we add

(C+) usually free of microorganisms.

Q) You find colorless area in bacterial cell in a Gram – stained smear. what should you do next:

- (C) an acid - fast staining
- (C) A flagella staining
- (C) A capsule staining

(C+) an endospore staining

- (C) a simple staining
-

Q) Which one of the followings is used in a negative bacterial staining because it's an acidic dye:

- (C) Pelican 2001

(C+) Nigrosine

- (C) 20% CuSo4
- (C) picric acid

Q) Heat is applied to all the following staining technique EXCEPT:

(C+) Negative staining

- (C) Direct simple staining
 - (C) Gram staining
 - (C) Acid Fast staining
 - (C) Endospore staining
-

Q) Which of the following staining techniques is used to study the natural size of bacteria?

- (C) Gram staining
- (C) Acid-fast staining
- (C) Endospore staining
- (C) Direct simple staining

(C+) Background staining

Q) Aseptic technique involves which of the followings?

- (C) Closing the air current of the lab
- (C) Disinfection of the table tops
- (C) Flaming of the inoculating loop
- (C) Working within a one-foot radius of the flame

(C+) All choices are associated with aseptic techniques

Q) Which one of the following concentration's of ethyl alcohol is the most effective decolorizing agent?

- (C) 50%
- (C) 30%
- (C) 70%
- (C) 40%

(C+) 95%

Q) the proper order for the steps of preparation of a smear of bacteria on microscope slide for the simple staining is :

- 1- clean the slide and place a small drop of sterile distilled water on the slide
- 2- allow the slide to dry at room temperature
- 3- use a sterile inoculating loop to spread bacteria on the slide
- 4- Quickly pass the slide through the flame to heat fix the cells to the glass.
- 5- Use a sterile inoculating loop to add bacteria to the drop of sterile distilled water

(C) 1-2-3-4-5

(C) 1-3-5-2-4

(C+) 1-5-3-2-4

(C) 1-5-3-4-2

(C) 1-5-2-3-4

Q) In the Direct simple staining, all the following are correct EXCEPT:

- (C) Direct simple staining uses only one stain
- (C) Direct simple staining is used for the Identification of the morphology (shape) of bacteria
- (C) Direct simple staining is used for the identification of the arrangement of bacterial cells

(C+) In the direct simple staining Gram positive, Gram negative and Acid-fast bacteria will be stained with the same color

(C) There is no decolorizing step in the simple staining

Q) Direct simple staining:

(C+) Increases the contrast between the cell and the background

- (C) Allows visualization of bacterial motility
- (C) Requires multiple staining steps
- (C) Differentiates one bacterial group from another
- (C) Requires one stain and one decolorizing agent

Q) The type of microscopy we use to examine bacteria that are stained by the Gram staining procedure is called:

(C+) Bright field microscopy

- (C) Fluorescent microscopy
 - (C) Dark field microscopy.
 - (C) transmission electron microscopy
 - (C) Phase contrast microscopy
-

Q) The procedure that kills cells and binds them to the glass slide is called:

(C) Smear preparation

(C+) Heat fixation

- (C) Hanging drop
 - (C) wet mount
-

Q) Why was it NOT advisable to apply heat to the slides during the negative staining procedure because:

(C+) Heat evaporate water from the bacterial cells and appear smaller than in nature

- (C) Most bacteria seen with negative staining do not need to be stuck to the slide
 - (C) Many bacteria cell are fragile and can't withstand applying heat
 - (C) Heat make the bacterial cells larger than in nature
-

Q) Which one the followings are correct about the size of the microscopic field

- (C) Remains constant regardless of the magnification of the objectives you are using
- (C+) Decrease when the magnification of the objectives you are used increase.**
- (C) increase when the magnification of the objectives you are used increase
- (C) you can control it manually regardless of the magnification of the objectives you are using
- (C) The information is insufficient.

Q) A non-contaminated culture is:

- (C) The observable growth that appears in or on the medium
- (C) Holds two or more identified microorganisms
- (C) Free of other living things except the one being studied
- (C) medium that grows only a single known species of microorganism

(C+) was once that contains different types of bacteria from the same source

Q) Bacterial cells can be determined by using which one of the following objective lens:

- (C) 40X
- (C) 10X
- (C) 4X

(C+) oil immersion lens

Q) You can increase the magnification of compound microscope by which one of the followings

- (C) adjusting of light intensity

(C+) changing the objectives lens you used

- (C) adjusting the specimen at the position
 - (C) changing the ocular lens you used
-

Q) what was one of the first and most useful microscopic tests for classifying bacteria that is still important today :

- (C) flagella staining
- (C) simple staining

(C+) Gram staining

- (C) negative staining for capsule

Q) In Gram staining the washing step after addition of primary stain is followed by:

(C) Loading the slide with crystal violet

(C) Loading the slide with safranin

(C+) Loading the slide with iodine

(C) Loading the slide with methylene blue

(C) Decolorizing step

Q) The decolorizing agent in Gram staining is:

(C) 3% Acid alcohol

(C) 70% alcohol

(C) H₂O

(C+) Absolute alcohol

Q) The bacteria which retained the color of primary stain in Gram staining will make which one of the followings with 10% KOH

(C) A mucoid mixture

(C+) non-mucoid mixture

(C) Depends on the type of bacteria

(C) The information is insufficient

Q) A Gram staining was carried for both Gram positive and Gram-negative bacteria using safranin as primary stain and crystal violet as a counter one. the color of these bacteria at the end of this staining technique. respectively, are:

(C) Pink, Violet

(C) Violet, pink

(C) Colorless, pink

(C) Pink, colorless

(C+) Both them violet

Q) At the same time, you prepared three bacterial slide from a 22 hours old pure culture through Gram staining which one of the followings explain the variability between these slides if you use the same Gram staining solution set

- (C) Age of the culture used
- (C) Freshness of the reagents.
- (C) Concentration of the decolorizing agent.

(C+) Thoroughness of washing by tap water

- (C) Type of the bacteria
-

Q) You believe you have isolated species to Bacillus and an isolated colony is on a plate which was incubated for two days. What portion of the colony would you use to get a correct classification of this bacteria through Gram staining?

- (C) both center and edge of the colony can be used
- (C) center of the colony
- (C) the part which is in between the edge and the center of the colony
- (C) All of the choices are incorrect

(C+) Edge of the colony

Q) The microscope part that is responsible for focusing the light up on the slide:

- (C) ocular lens
- (C+) Condenser**
- (C) Iris diaphragm
- (C) Objective lens
- (C) Fine focusing

Q) A microscope was pulled out of the storage cabinet.

Which one of the followings is a mistake was made in storing it?

(C) The scope was put in the case forwards, the arm facing out

(C+) A slide was left on the stage

(C) The stage at the lowest point

(C) Low power objective at the position

(C) The microscope was covered by dust cover

Q) Cleaning the lenses of the microscope is performed by using:

(C) Filter paper

(C) Tissue paper

(C) Cotton

(C) Alcohol

(C+) Lens paper.

Q) You are looking at bacteria with a compound microscope.

The ocular lens has a magnification or 8x and the objective lens has a magnification of 2x. What is the total magnification?

(C) 10X

(C) 4X

(C) 8X

(C) 10X

(C+) 16X

Q) Which of the followings is used in a negative staining because of its Large molecular size:

(C+) India ink

(C) 20% CuSo₄

(C) Nigrosine

(C) Carbol fuchsine

(C) Picric acid

Q) Bacterial cells are generally -----charged and thus repel the -----dyes:

- (C) Negatively, basic
- (C) Positively, acidic
- (C) Positively, basic

(C+) Negatively, acidic

- (C) Not, No

Q) Why are virtually all metabiological dyes are basic dyes?

- (C) Because the basic nature of the molecule is needed for the molecule to penetrate the cell
- (C) Because basic dyes are needed to neutralize the amino and nucleic acids
- (C) Because no other types of dye will bind to the capsule of the cell

(C+) Because the bacterial cell has an excess of negatively charged groups and basic dyes contain positively charged ionizable groups

- (C) Because the bacterial cell had an excess of positively charged groups and basic dyes contain negatively charged ionizable groups

Q) If the chromogen of stain is negatively charged, the stain is classified as a(n)

- (C) Basic stain.
- (C) Acid-fast stain
- (C) Cationic stain
- (C) Direct stain

(C+) acid stain

Q) All the following are basic stains EXCEPT :

(C+) Nigrosin

- (C) Safranin
- (C) Crystal Violet.
- (C) Methylene Blue
- (C) Carbol Fuchsin

Q) which of the following give the chromogen its charge in the stain:

(C) chromophore

(C+) Auxochrome

(C) Dye

(C) chromogen

Q) Aseptic technique involves all the following EXCEPT:

(C) disinfection of the table top

(C) flaming of the inoculating loop

(C) working within a one -foot radius of the flame

(C+) flaming of pipettes

Please forgive us for any mistake!

Healing group work team

Nursing group



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